Lec_44

Gate Questions on Quick Sort

Imp concept

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1.Consider the following sequence of i/p of an array

a) 1, 2, 3, 4,5 ,6 .....n

b)n ,n-1,n-2,n-3........1

c)n , n , n , n , n,......n, n, n times

5, 5, 5, 5, 5

If number of comparisons to arrange above sequence in ascending order using Quick sort are c1,c2,c3 than relation between c1,c2,c3 is ?

c1 = c2 = c3

$\frac{n(n-1)}{2}$

TC = $O(n^2)$

a, b, c → sort → using quick sort, than

c1, c2, c3 → ?

a) 1, 2, 3, 4, 5 - - - n → already sorted → QS = $O(n^2)$

1, 2, 3, 4, 5, 6, 7, 8

7 + 6 + 5 + - - - 1 → $\frac{7 \times 6}{2}$

n → $\frac{n \times (n+1)}{2}$

n-1 elements

no. of comparisons → $\frac{n(n-1)}{2}$ = $O(n^2)$

1, 2, 3, 4, 5, 6 - - - n

$c_1 = \frac{n(n-1)}{2}$

(b) ⇒ $n, n-1, n-2, n-3, n-4, \ldots, 1$

8, 7, 6, 5, 4, 3, 2, 1

i j

1, 7, 6, 5, 4, 3, 2, 8

$\Rightarrow 7 + 6 + 5 \ldots$

$\frac{7+6}{2}$

$\frac{n(n-1)}{2}$

$C_2 = \frac{n(n-1)}{2}$

$TC = O(n^2)$

(c) $n, n, n, n, \ldots$ upto n

S_a, S_b, S_c, S_d, S_e

$C_3 = \frac{n(n-1)}{2}$

a b c

c1 c2 c3

$C_1 = C_2 = C_3$

→ If array is already sorted in any order or all elements are same than number of comparisons, with using quick sort sorting technique are $\frac{n(n-1)}{2}$

$TC = \underline{O(n^2)}$

Imp point: Generally pivot element is choose first element if we randomly choose pivot than is called randomized quick sort

2.) in QS the sorting of n elements , n/5th element is selected as pivot using o(n^2) time than worst case time Complexity of Quick Sort is ?

$\Rightarrow O(n^3)$

$T(n) = T(n-1) + n$

$T(n) = 2T(n/2) + n$

BC

WC

partition (a, l, h)

$T(n) = T(n-1) + n + n^2$

$T(n) = T(n-1) + n^2$

$T(n) = T(n-1) + n^2$

$T(n) = T(n-2) + n^2 + (n-1)^2$

$T(n) = T(n-3) + n^2 + (n-1)^2 + (n-2)^2 \cdots 1^2$

$O(n^2)$

$1^2 + 2^2 + 3^2 - \ldots - (n-1)^2 + n^2$

$\frac{n(2n+1)(n+1)}{6} \Rightarrow O(n^3)$

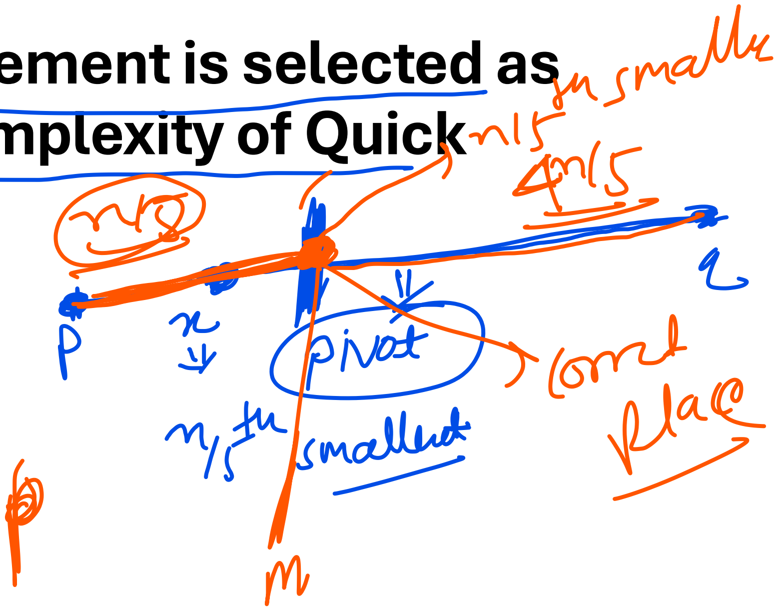
3.)in QS the sorting of n elements ,n/5th Smallest element is selected as pivot using o(logn) time than worst case time Complexity of Quick Sort is ?

$$T(n) = T(n/5) + T(4n/5) + n + \log n$$

$$\boxed{T(n) = T(n/5) + T(4n/5) + n}$$

$$\Downarrow$$

$$n\log n$$



4.)in QS the sorting of n elements , 300th largest element is selected as pivot using o(n^2) time than worst case time Complexity of Quick Sort is ?

10, 6, 9, 100, 32, 2 - - n

$T(n) = T(n-300) + T(299) + n + n^2$

change_pivot

partition

[10, 5, 6, 7, 3, 2]

n-300

$T(n) = T(n-300) + n^2$

smallest

299

300th le

$T(n) = T(n-1) + n$

$T(n) = T(n-300) + n^2$

ST

$O(n^3)$

$n^3/300$

$\rightarrow O(n^3)$

last question

4.)in QS the sorting of n elements , Median of array is selected as pivot than worst case time Complexity of Quick Sort is ?

$O(n)$

concept: To find median of an array mean is an algorithm (median of 5 algorithm) which takes $O(n)$ time.

n/2 n/2

$$T(n) = T(n/2) + T(n/2) + n + n$$

$$T(n) = 2T(n/2) + 2n$$

ans